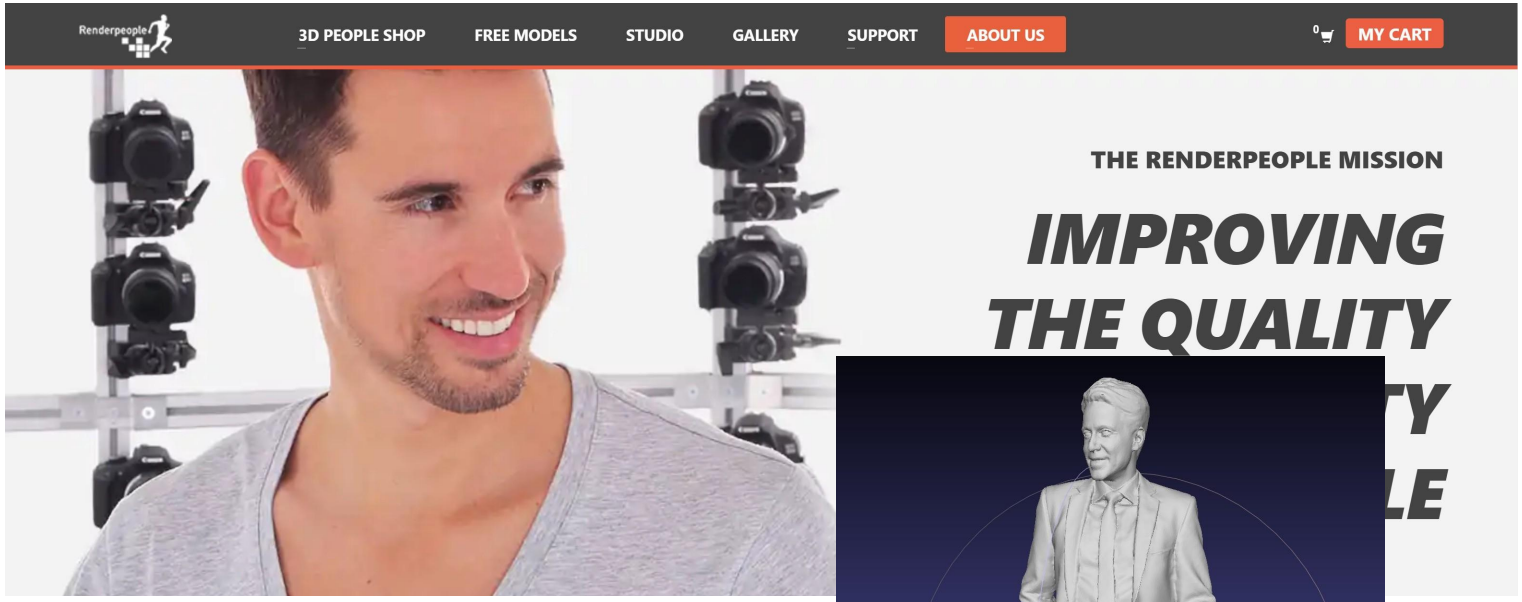


https://renderpeople.com/about-us/



Application: Virtual Reality Video



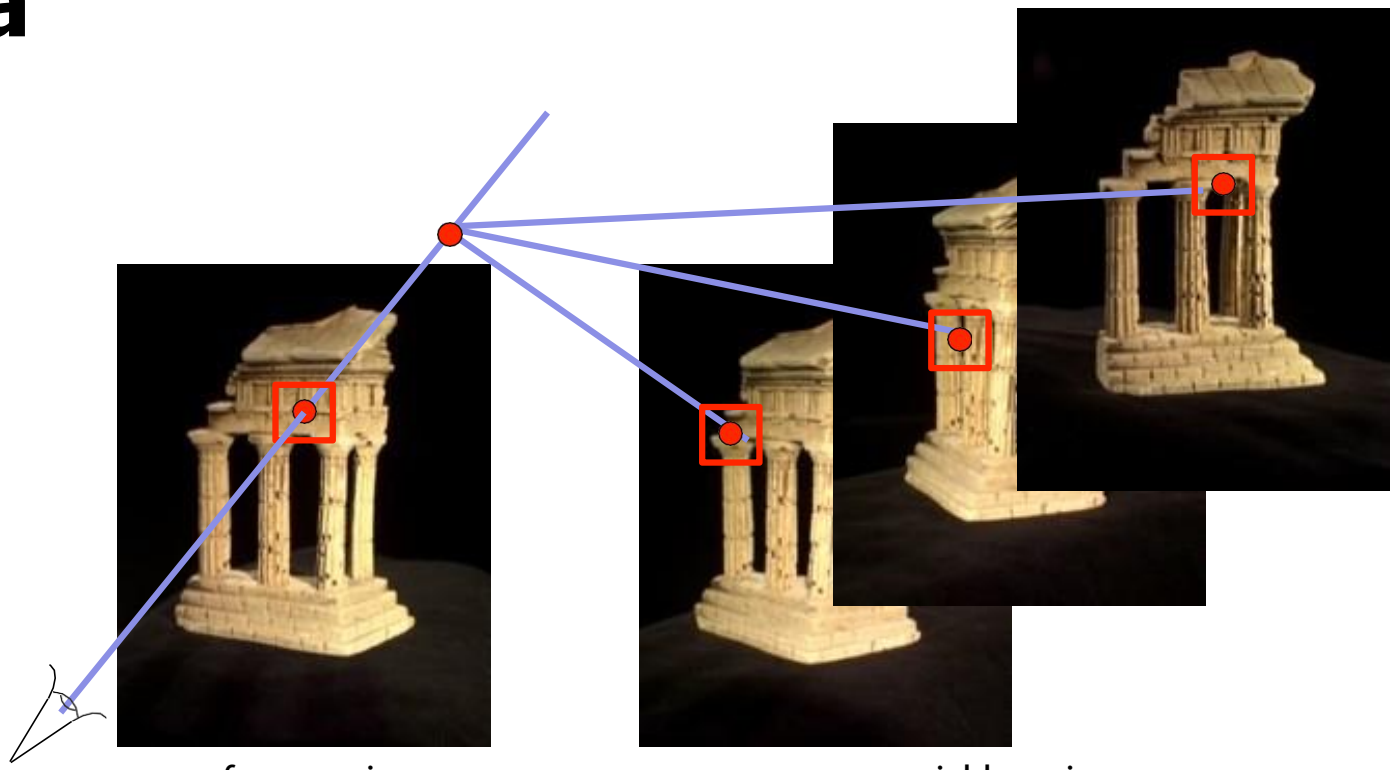
Anderson, et al. *Jump: Virtual Reality Video*. SIGGRAPH Asia 2016.



Broxton, et al. *Immersive Light Field Video with Layered Mesh Representation*. SIGGRAPH 2020



Multi-view stereo: Basic idea

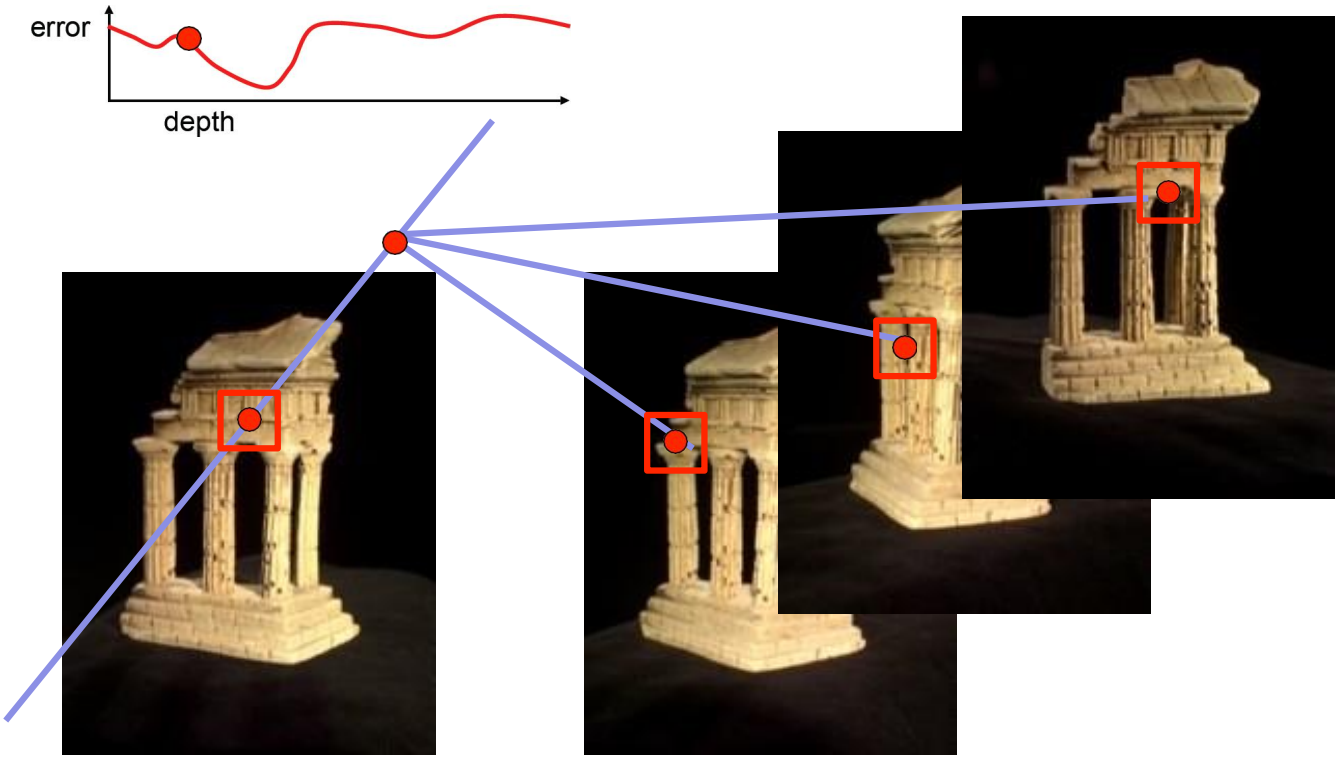


reference view

neighbor views

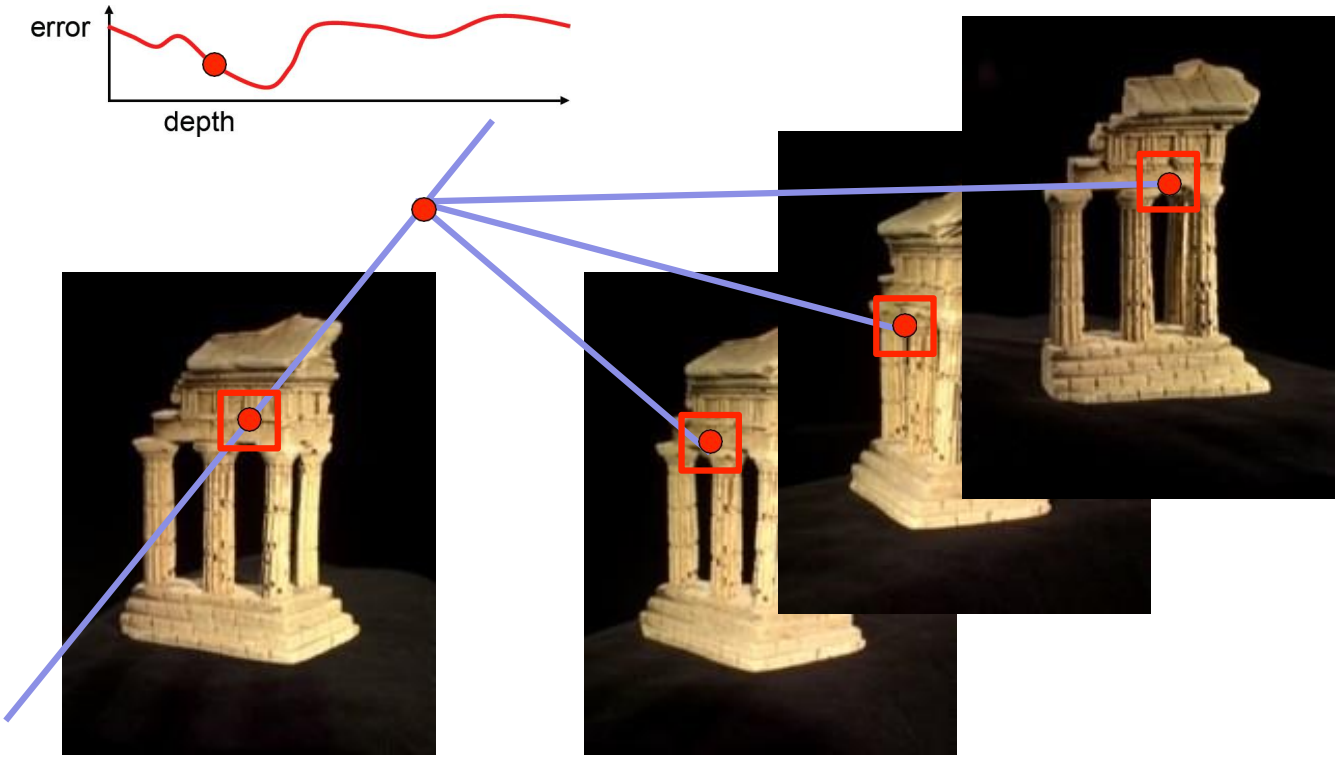
Source: Y.Furukawa

Multi-view stereo: Basic idea



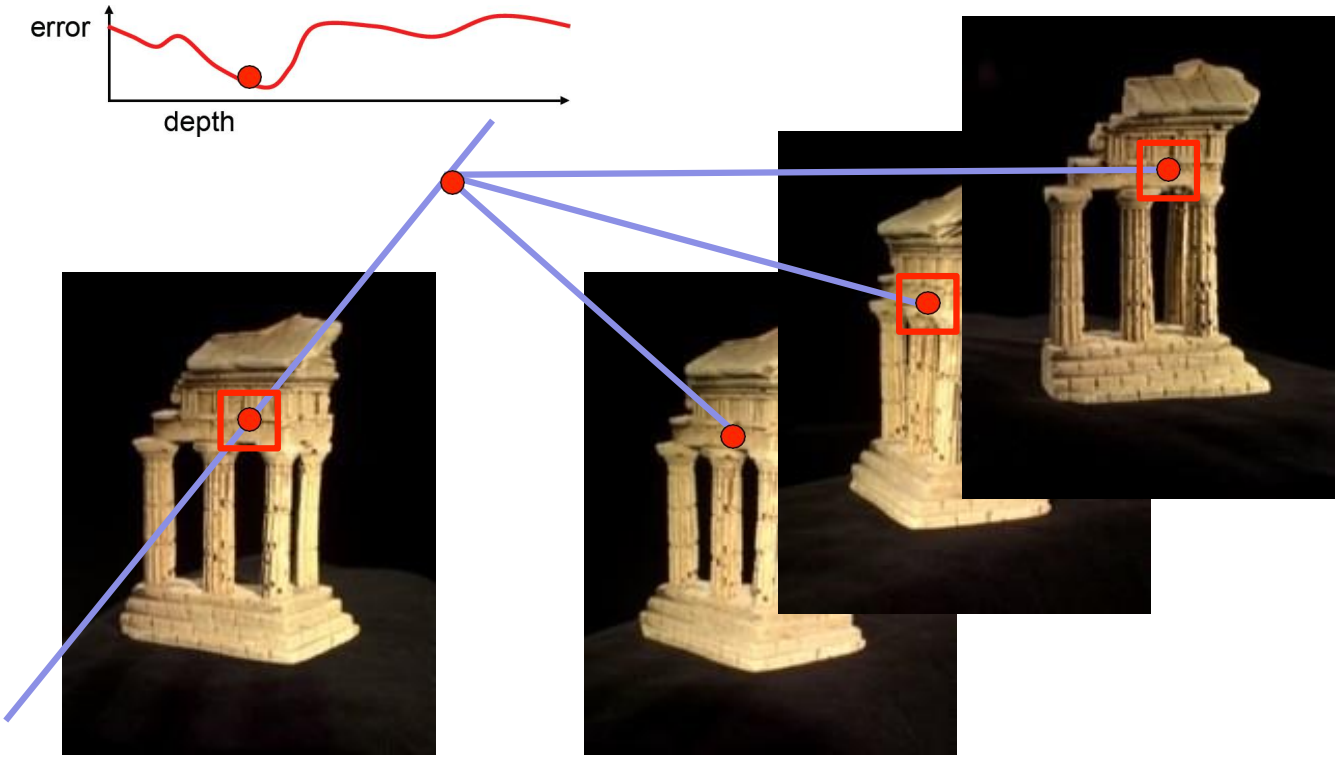
Source: Y.Furukawa

Multi-view stereo: Basic idea



Source: Y.Furukawa

Multi-view stereo: Basic idea



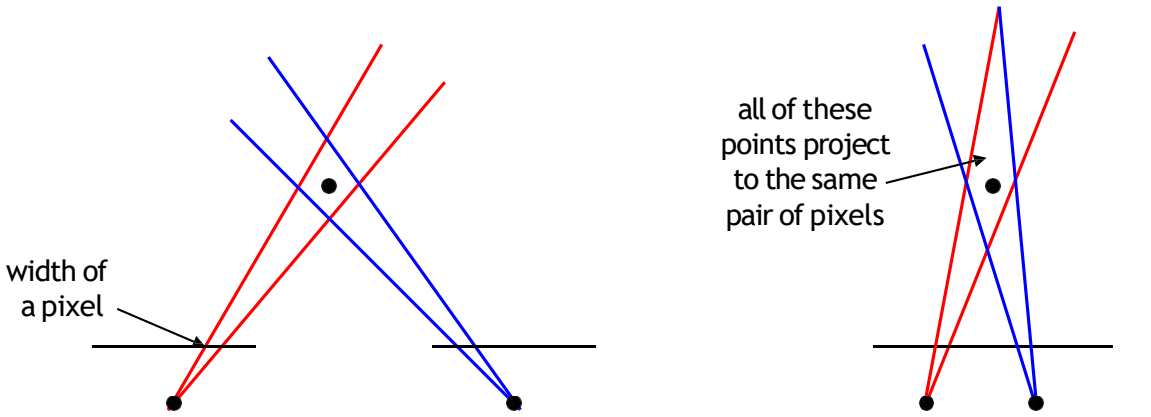
In this manner, solve for a depth map over the whole reference view

Source: Y.Furukawa

Multi-view stereo: advantages

- Can match windows using more than 1 neighbor, giving a **stronger match signal**
- If you have lots of potential neighbors, can **choose the best subset** of neighbors to match per reference image
- Can reconstruct a depth map for each reference frame, and the merge into a **complete 3D model**

Choosing a single stereo baseline



Large Baseline **Small Baseline**

What's the optimal baseline?

- Too small: large depth error
- Too large: difficult search problem